
On the statistical analysis of grouped data: when Pearson χ^2 and other divisible statistics are not goodness-of-fit tests

by Sara Algeri and Estate Khmaladze

R code description

The folder `Codes_Algeri_Khmaladze_JRSSB2025.zip` collects the R code needed to replicate the results presented in the manuscript *On the statistical analysis of grouped data: when Pearson χ^2 and other divisible statistics are not goodness-of-fit tests* by Sara Algeri and Estate Khmaladze. Below is a brief description of each file.

- File `Chandra_data.txt`: contains the data from the Chandra X-ray observatory described in Section 1.1.
- File `Chandra_analysis.R`: contains the R code used to conduct the analysis of the data from Chandra and discussed in Sections 4.1, 6.3.1, 7.1.1, and to reproduce Figure 1.
- File `Example_I.R`: contains the R code used for the power simulation for Example I described in Section 6.2 and to generate the left panel of Figure 2.
- File `Example_II.R`: contains the R code used for the power simulation for Example II described in Section 6.2 and to generate the central panel of Figure 2.
- File `Example_III.R`: contains the R code used for the power simulation for Example III described in Section 6.2 and to generate the right panel of Figure 2.
- File `Example_IV.R`: contains the R code used for the power simulation for Example IV described in Sections 6.4 and 7.1.
- File `Fig3.R`: contains the R code used to generate the left panel of Figure 3 in which the null distribution of the $\widehat{KS}$ and $\overline{KS}$ statistics are compared.
- File `Numerical_study_Sec_7_2.R`: contains the R code used to compare the CPU time required by the projected parametric bootstrap and the classical parametric bootstrap as described in Section 7.2.
- File `Sec_7_3_correction_term.R`: contains the R code used to derive the $\frac{0.6}{\sqrt{K}}$ shift for the KS^* statistic as described in Section 7.3.
- File `Fig4.R`: contains the R code used to generate the plots in Figure 4 in which the simulated null distribution of the $\overline{KS}$ and KS^* statistics are compared with the Kolmogorov distribution when testing different models.

For any question please email Sara Algeri at salgeri@umn.edu.